

1.Quadratic Equations

Polynomials Relations between roots and coefficient Formation of quadratic equations with given roots
Relations between roots and coefficient Nature of roots
Transformation of equations Common roots
Analysis of graph of quadratic Location of roots

2.Sequence and Series

Nth term of a sequence Arithmetic progression
Sum of terms in A.P. Arithmetic mean Geometric progression
Geometric progression Sum of terms in G.P.
Geometric mean Arithmetico-geometric progression Harmonic progression

3.Trigonometry

System of measuring an angle Regular polygon
Trigonometric functions
Compound angles formulae

4. Sets

Definition and description of sets Types of sets Subset & power set Operation on sets
Cardinality based problems

5.Relations& Functions

Cartesian product of sets Relation Inverse relation Definition & associated terms of a function
Classification of function Domain of a function
Modulus functions Greatest integer functions

6. Lines

Cartesian system of rectangular coordinates in a plane Section formula Centres of a triangle
Centres of a triangle Area of a triangle and collinearity
Line equation The slope of a line

7. Probability

Probability of an event, Probability of non occurrence of event
Classification of pack of cards
factorial notation and ncr notation and npr notation
Problems based on ncr and npr notation

8. Logarithms

Conversion of exponential form into logarithmic form
Addition and subtraction rule of logarithm

9. Statistics

Types of Data, mean of grouped data
Mean, Mode and Median
Standard Deviation and Variance

